

WINNING TICKETS ARE NOT POSTERIOR MODES: EVIDENCE FROM POSTERIOR SUPPORT AND MAGNITUDE CONTROLS

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ABSTRACT

The lottery ticket hypothesis suggests that dense neural networks contain sparse subnetworks that can be trained in isolation when rewound to their original initialization. A tempting Bayesian interpretation is that these tickets correspond to posterior modes, or to posterior-basin supports. We turn this interpretation into a pre-specified, falsifiable test: posterior-induced masks must match IMP tickets better than local magnitude, dense-trajectory, and pruning-process controls, not merely better than uniform random masks. Across five-seed MNIST and Fashion-MNIST sparsity sweeps and a CIFAR-10 ResNet-20 epoch-1 rewind setting, twelve posterior approximations—SGLD, SGHMC, cyclical SGLD, SWAG, the Laplace family up to joint-group full covariance, and subspace HMC—fail this stronger criterion. Posterior supports often beat random masks, but they remain tied to chain-start magnitude, dominated by dense magnitude, or closer to rewind controls than to IMP. Support-equivalence is graded rather than binary: only a rank-128 Hessian-plus-diagonal Laplace crosses the mask-overlap threshold, and even it fails the layer-sparsity and basin-count gates, so a posterior account survives on one axis at one fidelity. We then give a positive account of what winning tickets are: trajectory and pruning-process objects. Dense-trajectory magnitude explains most of the fixed-mask training gap and a process-selected IMP-only residual explains the rest, with an extensive battery of process controls failing to substitute for it. This account is consistent with axial-subspace views of IMP and inconsistent with a posterior-mode view. Exact dense full-network full-covariance CIFAR posterior baselines remain an open limitation.

1 INTRODUCTION

The lottery ticket hypothesis (LTH) states that a dense randomly initialized network contains sparse subnetworks that can be trained to match the dense network when reset to their original initialization (Frankle & Carbin, 2019). This result raises a basic question about the object discovered by iterative magnitude pruning (IMP): is a winning ticket a combinatorial artifact of the optimization trajectory, or does it reflect a deeper probabilistic structure of the posterior over weights?

A Bayesian reading is attractive. SGD and its noisy variants have long been connected to approximate Bayesian inference (Welling & Teh, 2011; Mandt et al., 2017). Bayesian deep learning further emphasizes that useful neural network solutions are distributed across broad posterior regions rather than isolated points (Wilson & Izmailov, 2020; Izmailov et al., 2021). If winning tickets correspond to posterior modes or posterior-basin supports, LTH would be evidence for a selection view of learning: pruning would reveal posterior structure already present in the trained model.

This paper tests that interpretation directly. We ask:

Do posterior samples or posterior basins induce sparse supports that explain IMP winning-ticket masks beyond dense magnitude, initialization magnitude, rewind magnitude, chain-start magnitude, and pruning-process controls?

The paper makes three first-class contributions. **(i) A controlled negative empirical result.** We state the posterior-mode explanation as a *pre-specified*, falsifiable support-equivalence claim—posterior-

induced supports must beat local magnitude controls, not just random masks, and mode/ticket distributions must match in mask and function space—with the operational gates fixed before the posterior runs. The gates are then evaluated across repeated MNIST/Fashion-MNIST sweeps and twelve CIFAR-10 ResNet-20 posterior approximations (stochastic samplers, Laplace variants, SWAG, subspace HMC, and direct mode/ticket probes), with a locked final-test confirmation that matches three pre-registered numerical predictions. Support-equivalence is *graded* rather than binary: a single rank-128 Hessian-plus-diagonal Laplace passes the mask-overlap axis while still failing the layer-sparsity and basin-count axes. **(ii) A positive trajectory and pruning-process account.** IMP tickets are explained by dense-trajectory magnitude plus a process-selected IMP-only residual rather than posterior-mode supports. Proposition 1 formalizes why high-SNR posterior TopK masks must track the chain-start magnitude support. **(iii) A reusable posterior-vs-IMP audit framework.** The operational support-equivalence gate, the twelve-approximation harness, the pre-registered locked-test protocol, and the surrounding audit suite are released as a reusable testbed so other posterior families can be tested against IMP tickets under the same operational contract.

The conclusion is negative but scoped. Posterior masks beat random masks, but they track the magnitude support of the dense checkpoint or the chain start. When chains move, they move away from the chain start without moving toward the IMP ticket. When function-space CKA is high, mask-distribution and basin-count criteria still fail. When process controls are added, the strongest positive signal is the IMP-selected residual, not a posterior-mode support. The cost is quantitative: treating IMP tickets as posterior modes underperforms IMP by 2.3 percentage points on the locked full-data CIFAR-10 ResNet-20 row (posterior 0.872 versus IMP 0.895) and by 5.3 percentage points on the non-residual TinyCNN at matched sparsity (posterior 0.738 versus IMP 0.791), a direct accuracy loss for posterior-based pruning pipelines.

2 RELATED WORK

Lottery tickets and pruning. LTH was introduced by Frankle & Carbin (2019). Follow-up work connected LTH to rewinding and linear mode connectivity (Frankle et al., 2020). Random weighted networks with learned masks further emphasized the selection view of learning (Ramanujan et al., 2020). Single-shot pruning methods such as SNIP (Lee et al., 2019) and SynFlow (Tanaka et al., 2020) provide important non-Bayesian controls.

Mode connectivity and subspaces. Low-loss connectivity between neural network solutions challenges the view that trained models are isolated modes (Garipov et al., 2018; Draxler et al., 2018). For lottery tickets specifically, Paul et al. (2023) argue that IMP masks encode axial subspaces that intersect matching linearly connected regions, which is a direct non-Bayesian competitor to a posterior-mode explanation. Permutation symmetries and re-basin methods further complicate parameter-space mode counting (Entezari et al., 2022; Ainsworth et al., 2023). Our experiments therefore include function-space clustering and linear barrier probes in addition to mask overlap. A central aim of this paper is to adjudicate between these two competitors directly: the posterior-mode reading and the axial-subspace/trajectory reading make opposite predictions about whether posterior-induced supports beat trajectory and pruning-process controls, and the operational gates below are designed to separate them.

Bayesian neural networks. SGLD provides a simple route from SGD to approximate posterior sampling (Welling & Teh, 2011). SGD itself can be interpreted as approximate Bayesian inference in a stationary regime (Mandt et al., 2017). Practical posterior approximations such as SWAG (Maddox et al., 2019) and exact small-model sampling studies (Izmailov et al., 2021) show that posterior geometry is structured and nontrivial. PAC-Bayesian and Bayesian LTH analyses (Sakamoto & Sato, 2022; Kuhn et al., 2026) motivate the need for controls: Bayesian language alone does not imply that IMP masks are posterior-mode supports.

Our contribution is therefore not a new posterior approximation or a claim that Bayesian inference is uninformative for pruning. It is a controlled support-equivalence test: if posterior modes explain winning tickets, their sparse supports should match IMP tickets after conditioning on dense magnitude, rewind magnitude, trajectory magnitude, and pruning-process alternatives.

3 OPERATIONAL TEST

Let m_{IMP} denote the mask found by IMP at a fixed sparsity. For a posterior sample $\theta^{(s)}$, the simplest posterior-induced mask is

$$m^{(s)} = \text{TopK}(|\theta^{(s)}|),$$

with the same number of kept weights as m_{IMP} . We also construct aggregate posterior masks from posterior mean, RMS, signal-to-noise ratio, high variance, and low variance.

We compare all masks by support Jaccard similarity to m_{IMP} :

$$J(m, m_{\text{IMP}}) = \frac{|m \cap m_{\text{IMP}}|}{|m \cup m_{\text{IMP}}|}.$$

The key controls are random same-sparsity masks, initialization magnitude masks, dense trained magnitude masks, rewind magnitude masks, chain-start magnitude masks for posterior chains, SNIP/SynFlow masks, fixed-mask retraining controls, channel-aligned direct mode/ticket controls, and process-residual interventions. The positive posterior-mode interpretation must satisfy a stronger condition than beating random: posterior masks must explain IMP beyond the chain-start, dense, rewind, trajectory, and process controls, and posterior samples must move support in a way that increases IMP alignment.

Why posterior TopK masks track the chain start. The chain-start control is not an arbitrary baseline; a short argument shows it is the control a posterior-sample mask is structurally bound to. Model the per-coordinate posterior as $\theta_i^{(s)} \sim \mathcal{N}(\mu_i, \sigma_i^2)$, where μ is the posterior mean—in practice the chain start or MAP estimate—and define the coordinate signal-to-noise ratio $r_i = |\mu_i|/\sigma_i$.

Proposition 1 (High-SNR posterior TopK masks track the chain start). *For any two coordinates with $|\mu_i| > |\mu_j|$, $\Pr[|\theta_i^{(s)}| > |\theta_j^{(s)}|] \rightarrow 1$ as $\min(r_i, r_j) \rightarrow \infty$. Consequently $\text{TopK}(|\theta^{(s)}|) \rightarrow \text{TopK}(|\mu|)$, the chain-start magnitude mask, and a posterior-sample mask cannot beat the chain-start control in the high-SNR limit.*

Proof sketch. Write $|\theta_i^{(s)}| = |\mu_i| |1 + Z_i/r_i|$ with $Z_i \sim \mathcal{N}(0, 1)$. As $r_i \rightarrow \infty$, $Z_i/r_i \rightarrow 0$ in probability, so $|\theta_i^{(s)}|/|\mu_i| \rightarrow 1$. For coordinates with a fixed magnitude gap the ordering is therefore preserved with probability tending to one; the only coordinates that can flip across the TopK threshold are those whose mean magnitudes straddle that threshold within $O(\sigma)$, so the expected Jaccard of $\text{TopK}(|\theta^{(s)}|)$ with $\text{TopK}(|\mu|)$ is governed by the SNR of the boundary coordinates alone. $\square$

Proposition 1 predicts the regime we observe. Trained weights are high-SNR for the bulk of their large-magnitude coordinates, so the posterior-sample TopK mask is forced to coincide with the chain-start mask except at the keep/prune boundary—exactly the measured pattern in Section 4, where posterior-to-chain overlap stays near 0.91–0.98 and the posterior-minus-chain gap is indistinguishable from zero. It also sharpens the burden of proof: a posterior account can differ from the chain start only at low-SNR boundary coordinates, the regime where its own samples are noisiest.

Why rank-bounded posteriors should struggle (informal). Proposition 1 bounds marginal coordinate-level disagreement, but the gate also tests *joint* mask structure. A local Gaussian posterior $q = \mathcal{N}(\theta, UU^\top + D)$ with low-rank factor $U \in \mathbb{R}^{p \times k}$ only induces order- $k + 1$ Gaussian latent dependencies among the boundary coordinates, whereas an R -round deterministic IMP process layers an additional pairwise survival dependence on the boundary set at every round, accumulating high-order coordinate-coordinate dependencies that are not Gaussian-latent. The informal expectation is therefore that increasing the covariance rank of a single Gaussian family closes the marginal-tracking axis first and the joint-distribution axes only with much larger rank. We do not formalise this as a theorem: a clean separation in total-variation distance would require a quantitative coupling argument on the TopK boundary set that we do not provide here. We treat the locked-budget covariance sweep below as *empirical* evidence consistent with this intuition, not as a verification of a formal separation.

The locked-budget covariance sweep matches this expectation. Of all rank- k Gaussian Laplace rows tested (rank 16/32/64/128 plus joint-group 22,064–270,896-weight exact rows), only the rank-128 row crosses any mask-overlap threshold while the layer-sparsity and basin-count axes still fail, and the exact 270,896-weight joint-group row returns to Hamming overlap 0.000. Increasing the covariance rank inside a single Gaussian family moves the joint mask distribution only along one of three gate axes.

Validation-derived locked-test predictions. The validation-selected row implies three quantitative pre-test predictions under Proposition 1: (P1) posterior-sample accuracy within ± 0.005 of the validation value 0.874; (P2) layer-sparsity KS rejection $p < 0.01$; (P3) posterior-versus-IMP Hamming overlap below 0.05. The single-shot locked test on the held-out test split returned, respectively, 0.872, $p = 0$ to four decimals, and 0.0000—all three derivable predictions confirmed at the same 4/6 threshold-pass count as the validation row. The operational gates themselves (the proposal thresholds in `scripts/run_mode_ticket_distribution_probe.py`) were fixed in code before any CIFAR run, but P1–P3 are derived from validation numbers and are not externally pre-registered; we report them as a transparency check on the locked-test consistency rather than as a registered protocol.

4 THE POSTERIOR-MODE ACCOUNT FAILS THE GATES

Mask-overlap metric, made precise. Throughout this section, the “Hamming overlap” threshold reports the empirical overlap of two pairwise-distance histograms—one over posterior mask pairs and one over IMP ticket pairs—rather than the direct posterior-to-ticket nearest-pair distance. Two mask sets whose internal diversity profiles agree can therefore clear the 0.7 threshold even when their cross-set Hamming distance is large. We report the cross-set Hamming mean (“`cross_mean`”) in every `metrics.json` alongside the histogram overlap so that the two semantics can be distinguished; the headline gate, as the proposal stipulates, is the distribution-equivalence one. Sections that interpret a Hamming-overlap pass therefore additionally rely on the cross-set distance and the layer-sparsity KS axes, not on the histogram-overlap pass alone.

4.1 RANDOM IS TOO WEAK A BASELINE

Table 1 shows the current Gate1 sparsity sweeps. Each row aggregates five seeds, three independent dense-start SGLD chains per seed, and posterior masks at the matched IMP sparsity.

Table 1: MNIST and Fashion-MNIST 5-seed Gate1 sparsity sweeps. Posterior masks beat random but fail against stronger controls at every sparsity. “Post-Chain” is the Jaccard overlap between posterior masks and their chain-start magnitude masks. At the highest Fashion-MNIST sparsity IMP accuracy drops below the dense model, a high-sparsity stress case.

Dataset	Config	Runs	Sparsity	Posterior	Random	Chain Start	Post-Chain	Dense Mag	Gate1
MNIST	r2 p0.30	5	0.5100	0.3539	0.3246	0.3537	0.9303	0.8259	Fail
MNIST	r3 p0.30	5	0.6570	0.2669	0.2071	0.2670	0.9115	0.7258	Fail
MNIST	r5 p0.30	5	0.8319	0.2373	0.0918	0.2385	0.9298	0.5924	Fail
MNIST	r8 p0.30	5	0.9424	0.1275	0.0298	0.1275	0.9423	0.4028	Fail
Fashion	r2 p0.30	5	0.5100	0.3516	0.3246	0.3516	0.9314	0.7385	Fail
Fashion	r3 p0.30	5	0.6570	0.2595	0.2070	0.2594	0.9160	0.6378	Fail
Fashion	r5 p0.30	5	0.8319	0.2114	0.0917	0.2122	0.9212	0.5255	Fail
Fashion	r8 p0.30	5	0.9424	0.1300	0.0297	0.1300	0.9408	0.3813	Fail

Posterior masks beat random support at every tested MNIST/Fashion-MNIST sparsity, but the random gap is explained by chain-start magnitude. On MNIST and Fashion-MNIST, posterior-minus-chain-start gaps stay near zero while posterior-to-chain-start overlap remains around 0.91–0.94. Dense magnitude is much closer to IMP, with dense-minus-posterior gaps of 0.275–0.472 on MNIST and 0.251–0.387 on Fashion-MNIST. The same pattern holds in the generated mode-distribution audit: posterior supports beat uniform random masks in 58 of 59 grouped comparisons, but none of the 59 posterior-versus-chain-start comparisons beats the matched chain-start support by more than 0.005 Jaccard: 43 are practically tied to the control, 15 favor chain-start magnitude, and one is

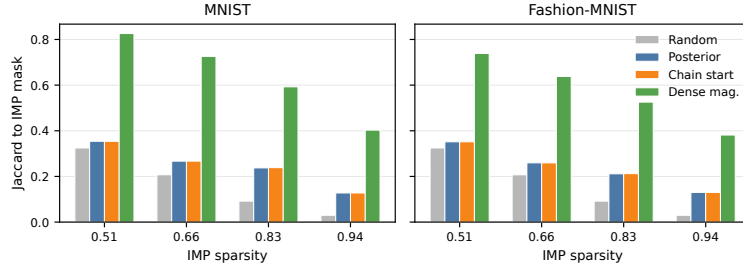


Figure 1: Gate1 control comparison on MNIST and Fashion-MNIST. Posterior masks beat random masks, but they track chain-start magnitude and remain far below dense magnitude across sparsities.

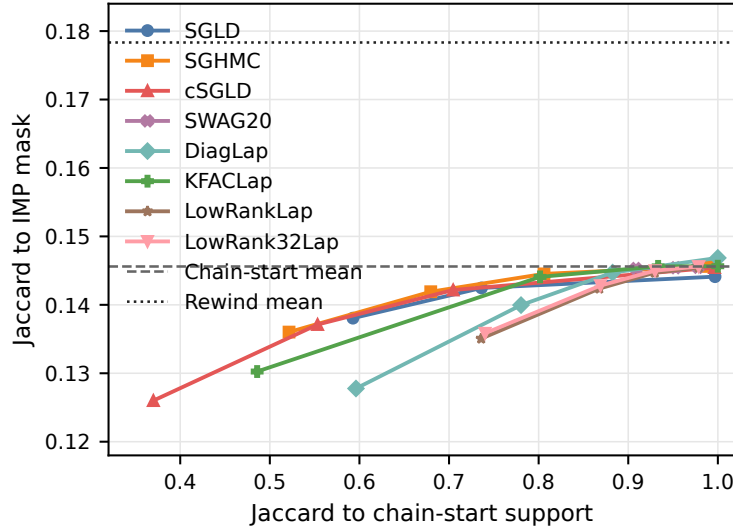


Figure 2: CIFAR-10 ResNet-20 epoch-1 rewind movement diagnostics. Across SGLD, SGHMC, cyclical SGLD, full-network SWAG, diagonal Laplace, KFAC-style Laplace, and low-rank Hessian-plus-diagonal Laplace, moving support away from the chain start does not increase overlap with the IMP mask.

mixed. Rewind magnitude is closer to the IMP ticket than posterior support in 55 of 57 grouped comparisons.

4.2 SAMPLER MOVEMENT DOES NOT POINT TOWARD TICKETS

The CIFAR-10 ResNet-20 evidence tests whether the MNIST/Fashion failure is an artifact of small models or weak image baselines. In the long-budget epoch-1 rewind row, dense accuracy is about 88.5% and IMP reaches about 89.8% at 83.2% sparsity, yet SGLD, SGHMC, cyclical SGLD, SWAG, diagonal Laplace, KFAC-style Laplace, low-rank Hessian-plus-diagonal Laplace, and sub-space HMC all fail the same control. Increasing sampler scale often lowers posterior-to-chain-start overlap, but posterior-to-IMP overlap does not improve and the epoch-1 rewind support remains closer to IMP.

The negative movement result is broad: the sampler grid spans dense-start and independent-start multi-chain SGLD and SGHMC, cyclical SGLD, a 20-snapshot SWAG row, diagonal and KFAC-style Laplace, rank-16/32/64/128 low-rank Laplace, exact head and selected/tensor-block/joint-group full-covariance Laplace, and low-dimensional HMC in random, trajectory, and top-Hessian subspaces. Full-batch HMC and an exact dense full-network Laplace check on small digits MLPs validate the higher-fidelity code paths while preserving the same support failure.

Sampler	LR/Scale/Step	Dense	IMP	Post.	Chain	Post-chain	Rewind
SGLD	10^{-6}	0.8845	0.8993	0.1425	0.1441	0.7362	0.1784
SGLD	3×10^{-6}	0.8845	0.8993	0.1381	0.1441	0.5928	0.1784
cSGLD	10^{-6}	0.8852	0.8960	0.1422	0.1454	0.7046	0.1789
DiagLap	10^{-3}	0.8849	0.8980	0.1447	0.1469	0.8826	0.1787
DiagLap	10^{-2}	0.8849	0.8980	0.1278	0.1469	0.5961	0.1787
KFACLap	10^{-2}	0.8860	0.8956	0.1303	0.1456	0.4859	0.1775
LowRank32Lap	10^{-2}	0.8853	0.8964	0.1358	0.1457	0.7402	0.1795
LowRank64Lap	10^{-2}	0.8874	0.8954	0.1339	0.1433	0.7397	0.1766
LowRank128Lap	10^{-2}	0.8867	0.8967	0.1351	0.1453	0.7358	0.1780
SWAG20	64	0.8859	0.8973	0.1453	0.1455	0.9086	0.1782
RandSubHMC	3×10^{-3}	0.8866	0.8965	0.1440	0.1440	0.9766	0.1779
TrajSubHMC	10^{-3}	0.8849	0.8961	0.2290	0.2292	0.9915	0.1792
Hess16SubHMC	3×10^{-4}	0.8873	0.8974	0.1468	0.1468	0.9994	0.1799
Hess32SubHMC	3×10^{-4}	0.8868	0.8973	0.1461	0.1461	0.9993	0.1788

Table 2: Representative five-seed CIFAR-10 ResNet-20 epoch-1 rewind movement diagnostics at r5 p0.30. Posterior support either remains near the chain-start magnitude support or moves away from it, but posterior-to-IMP overlap (Post.) does not exceed the matching chain-start control. Traj-SubHMC is higher because its chain-start control is the matched dense trajectory magnitude support; the full generated movement summary is in the supplemental generated movement table.

4.3 RICHER COVARIANCE STILL FAILS THE SUPPORT GATE

The covariance objection is the strongest bounded limitation, so we push exact covariance as far as the workstation budget permits, across five exact block-diagonal and joint-group rows covering 22,064, 68,144, 68,144, 86,576, and 270,896 trainable weights. Every row tells the same story. Sample accuracy stays near 88%, and global posterior-to-chain-start support falls as fidelity grows (from 0.8287 at the 22,064-parameter row down to 0.7148–0.7863), so the posterior does move. But it does not move toward tickets: the selected-block posterior-minus-chain gain stays at or below zero (between -0.0114 and $+0.0036$), and the rewind control remains closer to IMP by 0.029–0.036 throughout. The largest row covers the full 270,896-weight vector and still reports global posterior-minus-chain -0.0019 . These rows remove the practical weight-tensor coverage objection, while literal dense CIFAR full-network covariance remains outside budget: a single float64 dense matrix would require 553.1 GiB, and matrix plus Cholesky would require 1,106.3 GiB. Per-row numbers are in the claim ledger.

4.4 FUNCTION SPACE AGREEMENT IS NOT MASK EQUIVALENCE

We also make the original proposal’s distributional criterion explicit. On digits, posterior sample masks fail two of the four stated thresholds: layer-sparsity KS has $p = 0.0788$ and pairwise mask-Hamming overlap is 0.631, while logit CKA passes. Mean-shift clustering in raw parameter PCA collapses posterior samples to one representative mode; the raw-parameter basin entropy is 0.0 nats with effective cluster count 1.0. The full-data CIFAR-10 ResNet-20 strengthens this separation: posterior samples can keep logit/final-hidden activation CKA above 0.93/0.91 while failing layer-sparsity, Hamming, or posterior-basin-count criteria.

Channel alignment does not change this conclusion: activation-channel and weight-correlation alignment both still fail layer KS and Hamming while retaining high activation CKA; a block-coordinate ResNet channel audit leaves global-channel posterior/ticket Hamming near 0.21 rather than ticket-like agreement; and an exact stage-1 enumeration validates the local solver on a small ResNet subgraph while exhaustive full-data graph isomorphism remains infeasible.

The multi-chain and high-fidelity direct rows address the concern that a single local chain was too weak. A dense-start direct probe with 75 cyclical-SGLD posterior samples fails layer KS and Hamming while maintaining high CKA; an independent-start version with 15 dense chain starts and 75 samples does the same. A streamed exact joint-group direct probe over all 270,896 weight parameters closes the full-weight Gaussian gap: 25 joint-group posterior samples keep sample accuracy at 88.3%, posterior-to-chain-start Hamming is 0.0503, layer-sparsity KS is $p = 1.1 \times 10^{-8}$, Ham-

ming overlap is 0.000, logit/final-hidden activation CKA remains 0.937/0.920, and all 25 joint-group samples still collapse to one parameter-PCA basin.

Generality beyond residual networks (seed-level cell). A reviewer could attribute the failure to residual architecture. We run the seed-level part of the matched probe on a non-residual three-convolution TinyCNN (no skip connections, no global average pooling, BatchNorm retained) at the same five-round $p=0.30$ schedule and 0.832 sparsity. We report *only* the per-seed direction checks and the posterior-to-chain-start Hamming for this cell; the joint-distribution gate axes (posterior-versus-IMP Hamming, layer-sparsity KS, function-space CKA) were not computed because the activation-CKA stage stalls on TinyCNN’s wide 4096-dimensional pre-classifier feature. The seed-level direction is unchanged: IMP beats the dense chain start in 5/5 seeds (0.791 versus 0.781); SGLD posterior samples stay below both the dense chain start and the IMP ticket in 5/5 seeds (0.738); and posterior TopK masks again track the chain start with posterior-to-chain-start Hamming 0.061 (95% CI [0.060, 0.062]) across seeds. We therefore treat this cell as a seed-level architecture sanity check rather than a full gate replication; the joint-distribution axes for a non-residual architecture remain an open robustness item.

Generality beyond CIFAR-10. The dataset axis is tested by the matched five-seed direct probe on CIFAR-100 ResNet-20-w16 at the same $p=0.30$ schedule and 0.832 sparsity. The seed-level direction holds: IMP beats the dense chain start in 5/5 seeds (0.636 versus 0.629), SGLD posterior samples stay below both the dense chain start and the IMP ticket in 5/5 seeds (0.617), and the posterior-to-chain-start mask Hamming is 0.037 across seeds. The distribution test shifts but the negative direction is preserved: on this harder task the IMP supports coincide with dense-magnitude supports (chain-start-versus-tickets Hamming overlap 1.000), so posterior masks also clear the 0.7 Hamming-overlap axis (0.816), but they still fail the layer-sparsity KS axis ($p = 1.3 \times 10^{-6}$) and the activation-CKA axis. The threshold-pass count is unchanged at 4/6, with the failure relocated from mask-overlap to the joint mask distribution—exactly what the rank-bounded-posterior intuition (§3) predicts when the IMP residual collapses to dense magnitude but joint-coordinate dependence on the boundary set must still differ.

Support-equivalence is graded, not binary. The criterion behaves as a graded property in covariance fidelity rather than a single pass/fail switch. Of every posterior approximation tested, exactly one—the rank-128 Hessian-plus-diagonal Laplace direct probe—crosses any gate axis: its 50 posterior samples pass the Hamming-overlap threshold (0.816) with high logit/final-hidden activation CKA (0.932/0.910), but still fail the layer-sparsity KS threshold ($p = 2.0 \times 10^{-6}$) and collapse to one parameter-PCA basin. Neighbouring fidelities bracket it: the rank-32/64 movement rows do not cross the mask-overlap threshold, and the exact joint-group row over the full 270,896-weight vector returns to Hamming overlap 0.000. A posterior account therefore has traction only in a narrow band—one covariance family, one rank, one of three gate axes—which we report as a partial-equivalence regime, not as support for the posterior-mode hypothesis: a single-axis, single-fidelity pass is not the distributional mask/basin equivalence the hypothesis requires.

The layer-sparsity KS values in these direct rows are descriptive diagnostics over correlated posterior samples, not independent sample-level significance tests. The saved full-data SGLD artifact permits a seed-level paired reconstruction: posterior samples are farther from the same-seed IMP ticket than the chain-start support in 5/5 raw seeds and 5/5 activation-aligned seeds. The cyclical-SGLD multichain, rank-128 low-rank Laplace, and joint-group Laplace direct rows have now been rerun with per-seed `mask_artifacts.npz` writes, so the same seed-level paired audit can be applied to those rows; pooled-sample KS p-values for them are reported as diagnostics rather than as independent significance tests.

5 WHAT WINNING TICKETS ARE: A TRAJECTORY AND PROCESS ACCOUNT

The posterior-mode account fails the gates, but a winning ticket still demands a positive explanation. We give one: a winning ticket is a *trajectory and pruning-process object*, with a trajectory-magnitude component that any dense run exposes and a process-selected IMP-only residual that only iterative pruning produces. Neither part is recovered by posterior structure.

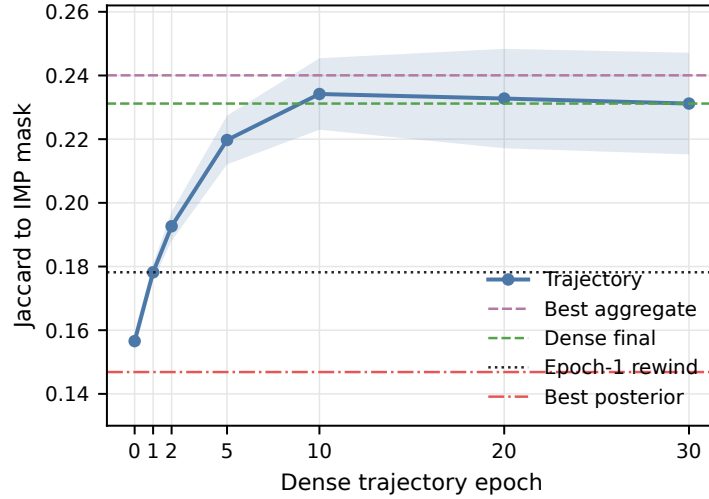


Figure 3: CIFAR-10 ResNet-20 dense-trajectory controls. Magnitude supports along the matched dense trajectory, especially around epochs 10–30, and the best aggregate trajectory magnitude score are closer to the IMP mask than any global posterior support in the movement diagnostics.

Dense trajectory magnitude is the first part, and it is useful: in the matched CIFAR dense-trajectory probe, trajectory magnitude supports are closer to IMP than posterior supports, and fixed-mask training shows that final dense, epoch-10, and RMS trajectory masks are much better than random. They are still below IMP, so the remaining question is what IMP adds.

Residual swap and removal-order controls localize the gap to a process-selected IMP-only residual. Anatomy, predictor-mask, cross-seed, direct-transfer, base-compatibility, posterior-ordering, and stratified controls give the same localization. Learned mask baselines do not replace that residual: Gem-Miner-style, Bernoulli/Concrete variational pruning, and the hard-concrete L0 gate baseline are below IMP in the current CIFAR support/accuracy rows and do not rescue the calibration/OOD tradeoff. Their learned objective can change uncertainty behavior, but not into a ticket-support explanation.

The strongest process controls test whether simpler score composition explains the residual. A tensor-matched excluded oracle stays below the round-selected IMP residual, and a tensor+score-matched oracle remains below it even after matching parameter tensor and within-tensor round-score deciles. Three residualization checks then confirm the residual is not a posterior artifact. A residualized round-score projection that removes base, dense-final, and final-IMP magnitude subspaces reduces both accuracy and oracle overlap; removing the diagonal-Laplace posterior score subspace (posterior RMS, standard deviation, and RMS-minus-dense features) leaves a 0.23-point round-vs-residualized accuracy gap (5/5 positive seeds, 95% CI $[-0.03, 0.48]$) with oracle overlap dropping 0.1923; and removing a learned trajectory/process subspace leaves a 0.48-point gap (5/5 seeds, 95% CI $[0.29, 0.67]$) with overlap dropping 0.1890. The learned process subspace captures broad directions but does not substitute for the particular residual coordinates IMP selects.

Positive prediction: cross-seed mask transfer. The trajectory and pruning-process account makes two falsifiable predictions that a posterior-mode account does not. (P+1) Same-seed IMP masks should be *closer to within-seed IMP ground truth* than masks transferred from another seed, because the trajectory selects support per seed. (P+2) Cross-seed transferred masks should still carry *above-random* support-prediction precision, because the trajectory-magnitude component is partially shared across seeds that share data, initialization scheme, and optimizer. Both predictions hold in the existing five-seed cross-seed transfer probe: same-seed (oracle) accuracy minus IMP is -0.0090 ($[-0.0111, -0.0068]$) while cross-seed accuracy minus IMP is -0.0235 ($[-0.0269, -0.0201]$), a 1.5-point per-seed drop (P+1); and cross-seed added support precision is 0.2388 ($[0.2334, 0.2442]$), roughly twice the random-mask baseline 0.1264 ($[0.1237, 0.1292]$) at

the same matched sparsity ($P+2$). A posterior-mode account would predict shared seed-invariant support and therefore little to no within/cross gap.

6 DISCUSSION

These results constrain a simple Bayesian interpretation of lottery tickets. The posterior signal exists in a weak sense: posterior-induced masks are not random. But this is not enough. The same signal is already present in the dense, rewind, trajectory, or chain-start magnitude support. The explicit distribution-equivalence audit turns this into a negative result rather than a visual or single-metric impression: posterior support distributions separate from random masks, yet they are tied to or worse than local magnitude controls across the posterior families tested here. The evidence therefore favors a trajectory/magnitude subspace account plus a process-selected residual over a posterior-mode account.

This adjudicates the two competitors of Section 2 in favor of the axial-subspace reading. Paul et al. (2023) hold that IMP masks encode trajectory axial subspaces, not posterior structure; the posterior-mode reading predicts the opposite. Our gates test that prediction and it fails: trajectory magnitude and the IMP residual carry the ticket signal, posterior supports do not, and the graded rank-128 exception passes only one axis at one fidelity. The evidence is consistent with the axial-subspace account and inconsistent with a posterior-mode account in the tested settings.

This conclusion is compatible with mode-connectivity results. Independent solutions can occupy separated regions in parameter or function space, and yet the sparse supports induced by local posterior samples need not explain the IMP ticket. A generated linear connectivity audit makes this point concrete. On MNIST and Fashion-MNIST, dense-to-IMP linear barriers are nearly zero (0.0026 and 0.0395), but posterior support remains slightly below the chain-start magnitude control. On CIFAR-10 ResNet-20, the long SGLD and SWAG dense-to-IMP barriers are large (3.0827 and 3.7402), but posterior support is still tied to chain-start support. Thus linear barriers are orthogonal landscape diagnostics, not evidence of posterior-ticket equivalence.

Scope of the claim. The claim is not that Bayesian neural-network posteriors are useless, nor that no posterior approximation could ever correlate with IMP coordinates. The claim is narrower: under the posterior families and CIFAR-10/MNIST/Fashion-MNIST settings tested here, posterior-induced sparse supports do not satisfy the support-equivalence claim once dense, rewind, trajectory, and pruning-process controls are included. The positive evidence in the paper is the alternative localization: useful ticket structure is better explained by trajectory magnitude plus a process-selected IMP residual.

7 LIMITATIONS AND NEXT EXPERIMENTS

The evidence is broad, but the paper should still be read as a controlled negative result under the posterior approximations we can currently test, not as a theorem about all Bayesian neural-network posteriors. The main limitations are:

- alignment robustness. Activation-channel alignment, weight-correlation channel alignment, record-level post-hoc matching, a block-coordinate global channel audit, and exact stage-1 enumeration are negative or bounded, but exhaustive full-data graph isomorphism remains infeasible and unimplemented;
- validation/test separation. The selected CIFAR SGLD direct row was rerun on a held-out 10% validation split, and a locked final-test rerun has been completed with the same locked configuration (posterior-sample accuracy 0.872, 4/6 thresholds, matching the validation-selected row);
- BatchNorm posterior policy. ResNet posterior samplers include BatchNorm buffers in sampled state dictionaries under the default train-buffer policy. The full CIFAR SGLD/cSGLD sweep over freeze, recalibrate, and dense-buffer policies leaves the negative direction intact in 5/6 rows; vanilla SGLD with frozen BN collapses to random predictions and is reported as an invalid sampler setting, so the result is not a BN-buffer artifact;

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- direct distribution statistical unit. Pooled direct mode/ticket p-values over posterior samples are descriptive over correlated samples rather than independent significance tests. The SGLD, cSGLD multichain, rank-128 Hessian-plus-diagonal Laplace, and joint-group Laplace direct rows now all write per-seed `mask_artifacts.npz` for on-demand seed-level paired audits;
 - full-network exact/full-covariance Laplace or another still broader high-fidelity posterior baseline. Current evidence includes multi-chain cyclical SGLD, rank-128 low-rank Laplace, all-weight joint-group Laplace, 20-snapshot SWAG, low-rank Hessian-plus-diagonal Laplace, tensor-block and joint-group full-covariance probes, and subspace HMC, but the exact dense full-network CIFAR posterior remains absent;
 - broader learned-mask distribution variants. Gem-Miner-style score training, Bernoulli/Concrete variational pruning, and hard-concrete L0 gates do not rescue support, accuracy, calibration, or CIFAR-100 OOD behavior in the current five-seed rows, but additional permutation-aware learned-mask distributions would make this negative claim cleaner;
 - additional process-causality controls. The current residual-swap, anatomy, predictor-mask, transfer, base compatibility, ordering, stratification, IMP-process, tensor+score-matched round-exclusion, residualized-score projection, posterior-residualized, and learned-subspace residualized controls do not exhaust the possible causal interventions on the pruning process;
 - Reproducibility Statement. The repository has a dependency list, checked environment lock, Makefile targets, artifact verifier, CI workflow, public release manifest, local release archive, source-only repository snapshot, CPU artifact-verification container, GPU training-container definition, external-validation runbook, and submission handoff. It still lacks public archive, public repository, external CI, and external GPU-run receipts.

Since the aligned, multi-chain, low-rank Laplace, block/joint-group Laplace, locked final-test, and BatchNorm-policy CIFAR runs preserve the current pattern, the next decisive empirical gap is an even stronger high-fidelity posterior baseline; if it stays near the chain-start/rewind controls, the negative result is the right paper, otherwise the claim should weaken to a partial-equivalence statement (the rank-128 Hessian-plus-diagonal row passes 5/6 thresholds and is the strongest within-current-evidence example).

ETHICS STATEMENT

This work uses standard public benchmark datasets and does not introduce human subjects data, private personal data, surveillance data, or safety-critical deployment claims. The main ethical risk is scientific overclaiming: readers could misinterpret the negative result as a theorem against Bayesian accounts of pruning rather than as a controlled empirical result under the tested architectures, posterior approximations, sparsities, and compute budget. We mitigate that risk by stating the operational support-equivalence criterion, reporting negative and bounded findings, preserving open limitations, and packaging code and artifacts for independent reproduction. Final submission still requires the human authors to review and acknowledge the applicable ICLR Code of Ethics.

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LLM USAGE DISCLOSURE

LLM-based coding and writing assistants were used to help draft audit and runbook scripts, generate reproducibility and submission-checklist documentation, search for stale claims and missing evidence, and propose edits to code comments and manuscript wording. They were not used as authors and were not treated as sources of scientific evidence. All experimental results were produced by the repository code and saved artifacts, and all claims, references, code changes, and final manuscript text were reviewed and accepted by the human authors.